A Joint Minimization Framework for Transformer Interpretability

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ABSTRACT

I present a joint minimization framework for the transformer interpretability problem. Given an observed output token $\hat{x}$, I seek the minimal subset $S \subseteq [d]$ of model parameters and $T \subseteq [n]$ of input positions sufficient to reproduce $\hat{x}$ under the same model. The central insight is that softmax does not affect argmax: the preservation condition reduces to a margin inequality $\delta > 0$ rather than full distribution matching, which is strictly weaker and more tractable. Under this relaxation, I (a) conjecture that T is identified by attention weight concentration rather than positional recency; (b) decompose S into a final-block component S_L and a residual-stream support set S_{supp} whose perturbation contribution to δ must be formally bounded; and (c) establish that S and T are not independently minimizable — a change in T reshapes the residual stream and therefore changes which S is minimal. I connect the sparsity conjecture to the Lottery Ticket Hypothesis [4], invoke ROME-style causal tracing [5] as partial empirical support, state the formal sufficient condition required to convert the conjectures into theorems, and outline a GPT-2 ablation protocol for empirical verification.

Keywords: *transformer interpretability, minimal circuits, input attribution, residual stream, logit margin, lottery ticket hypothesis*

1. INTRODUCTION

The interpretability of large language models (LLMs) is among the central open problems in AI safety and mechanistic understanding. Existing approaches — including input attribution [2, 3], sparse circuit identification [1], and causal tracing [5] — typically treat parameter minimization and input minimization as separate, independent tasks. This paper introduces a unified formulation that jointly minimizes both.

The key insight is elementary but consequential: softmax does not affect argmax. Therefore, the interpretability problem does not require preserving the full output distribution P_θ — it requires only that the winning logit remains the winner. This transforms a density-matching problem into a margin-preservation problem, which admits substantially sparser solutions.

A prior version of this framework conjectured that S and T could be minimized independently. This draft corrects that assumption and identifies the T-dependence of S^* as the primary open problem. The refined framework makes three contributions:

- **Softmax Relaxation.** I replace the full distribution-preservation condition with the strictly weaker margin condition $\delta > 0$, and show this makes small (S, T) pairs plausible.
- **Residual Stream Decomposition.** I decompose S into a final-block component S_L and a support set S_{supp}, and connect S_{supp} to the residual stream framework of Elhage et al. [1].
- **Coupling Problem.** I formally identify the dependence of S^* on T as the primary open problem, state the greedy decomposition as a tractable approximation, and provide the sufficient condition required for a complete theorem.

2. PROBLEM FORMULATION

Let V be a finite vocabulary and $X = (x_1, x_2, \ldots, x_n) \in V^n$ an input sequence. A language model parameterized by $\theta \in \mathbb{R}^d$ defines the conditional distribution:

$$P_\theta(x_{n+1} \mid x_1, \ldots, x_n) = \mathrm{softmax}(f_\theta(X)) \tag{1}$$

where $f_\theta : V^n \to \mathbb{R}^{|V|}$ is a composition of L non-linear transformations:

$$f_\theta = f_\theta^{(L)} \circ f_\theta^{(L-1)} \circ \cdots \circ f_\theta^{(1)} \tag{2}$$

The model's predicted token is:

$$\hat{x} = \mathrm{argmax}_{v \in V}\, P_\theta(v \mid X) \tag{3}$$

Problem Statement.

Given observed output token $\hat{x}$, find minimal $S \subseteq [d]$ and $T \subseteq [n]$ such that:

$$\mathrm{argmax}_{v \in V}\, P_{\theta_S}(v \mid X_T) = \hat{x}$$

where θ_S denotes θ restricted to parameter indices S, and X_T denotes X restricted to input positions T.

Since f_θ is non-linear and non-invertible, this problem is computationally intractable in the general case for $d \sim 10^{11}$. Section 3 shows how the intractability is substantially relaxed under a natural weakening of the preservation condition.

3. THE SOFTMAX RELAXATION

3.1 Margin Definition

The standard formulation implicitly requires preserving the full distribution P_θ. I replace this with a strictly weaker condition. Define the prediction margin δ as the gap between the winning logit and its nearest competitor:

$$\delta = f_\theta(X)[\hat{x}] - \max_{v \neq \hat{x}} f_\theta(X)[v] > 0 \tag{4}$$

The argmax condition holds if and only if $\delta > 0$. Since softmax is a monotone transformation, preserving the argmax requires only preserving the sign of δ — not any particular probability value. This is the key relaxation.

3.2 The δ-Buffer Argument

Most parameters in a language model do not affect the identity of the winning token: they refine the probabilities of losing tokens, smooth the output distribution, or encode features relevant to positions not in T. Formally, a parameter θ_i is irrelevant to $\hat{x}$ if perturbing it changes the logit margin δ by less than δ itself — that is, the margin remains positive after the perturbation. The set of irrelevant parameters is large precisely because $\delta > 0$ creates a buffer zone around the decision boundary. This observation motivates the existence of small (S, T) pairs.

3.3 Formal Sufficient Condition

The following condition, if proved, converts all conjectures in this paper into theorems:

Sufficient Condition (Open Problem).

Prove that for all $\theta_i \notin S$ (i.e., θ_i in the complement $S^c = [d] \setminus S$), removing θ_i changes the winning logit by less than δ. Formally:

$$|\mathrm{logit}_{\hat{x}}(\theta_S, X_T) - \mathrm{logit}_{\hat{x}}(\theta, X)| < \delta$$

A proof via Lipschitz continuity of f_θ or first-order perturbation analysis would suffice. The primary challenge is the non-linearity of f_θ, which makes tight analytical bounds difficult to derive.

Note: a first-order Taylor expansion gives $\Delta\mathrm{logit} \approx \nabla_\theta \mathrm{logit}^T \cdot \Delta\theta$, suggesting that parameters with small gradient magnitude are natural candidates for S^c (the complement of S, i.e., the set of removable parameters).

4. MINIMIZING T: INPUT POSITION SELECTION

4.1 Attention-Weight Formulation

I first define positional salience. Let $\alpha^{h,l}(t)$ denote the attention weight assigned to position t by head h at layer l, where h ranges over all attention heads and l ranges over all L layers. Define the cumulative salience of position t as:

$$\sigma(t) = \Sigma_{h,l} \alpha^{h,l}(t) \tag{5}$$

For a threshold $\varepsilon > 0$ chosen such that positions with $\sigma(t) < \varepsilon$ contribute less than $\delta/2$ to the winning logit, define the salient set:

$$T_sal = \{ t \in [n] : \sigma(t) > \varepsilon \} \tag{6}$$

I then conjecture that T decomposes as:

$$T = \{ x_{n-k}, \ldots, x_n \} \cup T_sal, \quad k \ll n \tag{7}$$

where the suffix captures the recency component and T_sal captures earlier positions with high semantic weight. Positions satisfying $\sigma(t) < \varepsilon$ contribute negligibly to the residual stream and can be removed without crossing the δ boundary.

4.2 Failure Modes of the Suffix Assumption

Transformers use global attention. A token at position 1 can dominate the logit landscape if it carries high semantic weight. This constitutes the principal failure mode of the T conjecture. Three concrete cases break the suffix assumption:

- **Negation at sentence onset:** *"Despite everything, the answer is still…"* — 'Despite' at position 1 reshapes the entire logit distribution, overriding any recency signal from the suffix.
- **Long-range syntactic dependencies:** The subject introduced at positions 1–3 may determine verb form or object type 20 or more tokens later, contributing decisively to $\hat{x}$.
- **Named entities:** *"Harry Potter picked up his…"* — the entity at positions 1–2 causes the model to predict 'wand' rather than 'phone'. The suffix alone is insufficient.

The corrected claim is: T is identified by attention weight concentration, not positional recency. The suffix approximation is useful in practice but cannot be taken as a structural guarantee.

5. MINIMIZING S: PARAMETER SELECTION

5.1 Residual Stream Decomposition

I conjecture that S decomposes into two components:

$$S = S_L \cup S_supp \tag{8}$$

where S_L denotes the parameters in the final transformer block that directly govern the logit margin δ, and S_supp denotes earlier-layer parameters whose representations are consumed by S_L via the residual stream [1].

Specifically, S_L consists of: (i) the unembedding projection weights for $\hat{x}$ and its nearest competitor; (ii) the attention heads in the last one or two layers that activate most strongly on T; and (iii) a small set of MLP neurons in the final transformer block.

S_supp is nonzero but empirically small. Its contribution to δ must satisfy the bound:

$$| \delta(S_L) - \delta(S_L \cup S_supp) | < \delta / 2 \tag{9}$$

If this bound holds, removing S_supp does not flip the argmax.

5.2 Connection to the Lottery Ticket Hypothesis

Frankle and Carlin [4] demonstrated that dense neural networks contain sparse subnetworks — 'winning tickets' — capable of matching the full network's performance when trained in isolation. My conjecture is the inference-time analogue: for a fixed input X and prediction $\hat{x}$, there exists a sparse subnetwork S that preserves the argmax without retraining. The Lottery Ticket result provides existence justification for small S at training time; the present framework conjectures that analogous sparsity holds at inference time for the argmax condition.

5.3 Middle-Layer Localization and ROME

Meng et al. [5] demonstrated via causal tracing that factual associations in GPT-class models are localized in middle-layer MLP modules. This directly challenges the assumption that S_L alone is sufficient: for factual recall queries such as 'The capital of France is…', the critical parameters may reside in middle layers rather than in the final block. This implies that S_{supp} may be substantially larger for factual queries than for syntactic continuation tasks, motivating stratification by query type in empirical verification.

6. JOINT MINIMIZATION: THE COUPLING PROBLEM

6.1 The Dependence of S* on T

S and T are not independently minimizable. When T changes, the token embeddings fed to layer 1 change. This perturbation propagates through all L layers via the residual stream, producing a different hidden state at layer L. The set of parameters in S_L that are load-bearing for δ therefore changes as a function of T. Formally:

$$S^*(T) = \operatorname{argmin} |S| \quad \text{s.t.} \quad \operatorname{argmax}_{v \in V} P_{\theta_S}(v \mid X_T) = \hat{x} \tag{10}$$

S* is a function of T. Treating S and T as independent is an approximation whose error is bounded only if S_{supp} is small relative to δ. The coupling through the residual stream constitutes the primary open problem of this framework.

> **Open Problem.**
>
> Characterize the function T → S*(T). Specifically:
>
> (1) Prove or bound |S*(T) − S*(θ, X)| as a function of |T|.
>
> (2) Determine whether the greedy decomposition (fix T, then minimize S | T) is within a constant factor of the true joint minimum.

6.2 Greedy Decomposition

Despite the coupling, a greedy decomposition provides a computable upper bound on |S| + |T|. The procedure runs in O(n) attention queries for T and O(d) gradient evaluations for S:

1. **Identify T** using cumulative attention weight analysis (Section 4). This minimizes |T| for the full parameter set θ.
2. **Given T, identify S** by gradient-sensitivity scoring of parameters with respect to the residual stream produced by X_T.
3. **Check coupling:** does S*(T) differ materially from S*(θ, X)? If the discrepancy is small, the greedy decomposition is valid. Otherwise iterate.

The greedy decomposition is suboptimal in general but tractable and produces empirically verifiable bounds. The GPT-2 ablation protocol in Section 7.2 uses this decomposition as its primary experimental design.

7. SUMMARY AND EMPIRICAL VERIFICATION

7.1 Summary of the Framework

Subset	Refined Conjecture	Failure Mode	Status	Key Bound
T (Input)	Suffix $\{x_{n-k},\ldots,x_n\} \cup T_sal$ (attention-identified; Eq. 6–7)	Global attention: position-1 token with high $\sigma(t)$ can dominate δ	Conjecture	$\Sigma_\{h,l\}\ \alpha^\{h,l\}(t) > \varepsilon$
S (Params)	$S = S_L \cup S_supp$; final block + residual stream writers	Factual queries: middle-layer MLPs (ROME [5]) may dominate S_supp	Conjecture	$\lvert\delta(S_L) - \delta(S_L \cup S_supp)\rvert < \delta/2$
Joint (S,T)	Greedy: fix T via attention, then minimize S | T	$S^*(T) \neq S^*(\theta, X)$ in general; gap uncharacterized	Open Problem	$S^*(T)$ is a function of T
Condition	$\delta > 0$ preserved under (S, T) restriction	No formal Lipschitz bound proved	Required	$\lvert\Delta\text{logit}\rvert < \delta$

Table 1. *Summary of conjectures, failure modes, open problems, and key bounds.*

7.2 GPT-2 Ablation Protocol

All claims in this paper require empirical verification on open models. GPT-2 [6] is suitable because its full parameter set and attention weights are accessible. I propose the following protocol:

1. **Sampling.** Sample 1,000 input sequences stratified by type: syntactic continuation, factual recall, negation-heavy, and long-range syntactic dependency.
2. **Baseline.** Record $\hat{x}$ for each sequence using the full model θ.
3. **T identification.** Greedily remove input positions in ascending order of $\sigma(t)$. Record the minimal T at which $\hat{x}$ first flips.
4. **S identification.** Greedily ablate parameters in ascending order of gradient-sensitivity score. Record the minimal S at which $\hat{x}$ first flips.
5. **Size measurement.** Measure $|S|/d$ and $|T|/n$ distributions per query type; report means, medians, and 90th percentiles.
6. **Coupling test.** For each T found in Step 3, recompute $S^*(T)$ and compare $|S^*(T)|$ against $|S^*(\theta, X)|$. A small discrepancy empirically validates the greedy decomposition.

8. LIMITATIONS

- **Suffix approximation is not a theorem.** T is identified by attention analysis per input. There is no structural guarantee that T is suffix-heavy.
- **S_supp is unbounded.** The inequality $|\delta(S_L) - \delta(S_L \cup S_supp)| < \delta/2$ is a requirement, not a proved result. For factual queries, S_supp may be large.
- **Greedy decomposition is suboptimal.** The true joint minimum of (S, T) may be smaller than the greedy bound. The gap is uncharacterized.
- **Sufficient condition unproved.** All conjectures depend on a Lipschitz or perturbation bound on f_θ that has not been derived.
- **Scale.** Verification on GPT-2 does not guarantee generalization to models with $d \sim 10^{11}$. Distributed representations in large models may yield large (S, T).

9. CONCLUSION

I have proposed a joint minimization framework for transformer interpretability grounded in a margin-preservation relaxation of the standard problem. The core observation — that softmax does not affect argmax, and therefore only the sign of δ needs to be preserved — substantially reduces the apparent intractability of the problem.

Under this relaxation, I conjecture that minimal (S, T) pairs are concentrated in the final transformer block and a small attention-salient input subset respectively. I have identified the T-dependence of S* as the central open problem, proposed a greedy decomposition as a tractable approximation, connected the sparsity conjecture to the Lottery Ticket Hypothesis, and invoked ROME-style causal tracing as partial empirical support.

The primary open problem is the formal sufficient condition: a perturbation bound proving that removing $\theta \notin S$ changes the winning logit by less than δ. Without this bound, the framework is a well-motivated conjecture. With it, it becomes a formal result.

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10. REFERENCES

[1] Elhage, N., Nanda, N., Olsson, C., et al. (2021). A Mathematical Framework for Transformer Circuits. Anthropic Technical Report.

[2] Lei, T., Barzilay, R., & Jaakkola, T. (2016). Rationalizing Neural Network Predictions. In Proceedings of EMNLP 2016.

[3] Bau, D., Belinkov, Y., Sajjad, H., et al. (2020). Identifying and Controlling Important Neurons in Neural Machine Translation. In Proceedings of ICLR 2020.

[4] Frankle, J. & Carlin, M. (2019). The Lottery Ticket Hypothesis: Finding Sparse, Trainable Neural Networks. In Proceedings of ICLR 2019.

[5] Meng, K., Bau, D., Andonian, A., & Belinkov, Y. (2022). Locating and Editing Factual Associations in GPT. NeurIPS 2022.

[6] Radford, A., Wu, J., Child, R., et al. (2019). Language Models are Unsupervised Multitask Learners. OpenAI Technical Report.